THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 5: **Exploring Forces** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A force is a: 1

(a) type of mass **(b)** kind of energy **(c)** push or pull **(d)** speed

2. Pressing an inflated balloon shows that force can change: 1

(a) density **(b)** colour **(c)** mass **(d)** shape

3. Force due to muscles is called: 1

(a) electrostatic force **(b)** magnetic force
(c) gravitational force **(d)** muscular force

4. Aeroplanes are streamlined to reduce: 1

(a) mass **(b)** weight
(c) gravity **(d)** friction due to air

5. A rubbed plastic scale attracts paper bits due to: 1

(a) gravity **(b)** friction
(c) electrostatic force **(d)** magnetic force

6. An object thrown upwards slows down because of: 1

(a) its own colour **(b)** air only
(c) gravitational force **(d)** magnetic force

7. Which quantity remains the same everywhere? 1

(a) mass **(b)** gravitational pull
(c) weight **(d)** force of friction

8. An instrument used to measure weight is a: 1

(a) spring balance **(b)** beam balance only
(c) barometer **(d)** thermometer

9. The upward force exerted by a liquid on an object is called: 1

(a) weight **(b)** friction
(c) pressure **(d)** upthrust (buoyant force)

10. Pumice floats on water because it: 1

(a) has trapped air and is less dense than water
(b) is very heavy
(c) is magnetic
(d) is a metal

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Define force and give its SI unit. 2

12. List four effects a force can have on an object. 2

13. Differentiate contact and non-contact forces with one example each. 2

14. What is muscular force? Give two examples of its use. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain with an example how a force can change the speed of an object. 3

16. Write the effects of force with one example each: start motion, change shape, change direction. 3

17. Why does an object at rest not necessarily have no force acting on it? 3

OR

Explain the origin of friction. 3

18. Give three ways in which friction is reduced in vehicles or ships and why. 3

OR

Describe an activity showing electrostatic force using a plastic scale and paper bits. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Define force. Describe four effects of a force with an example of each. 5

OR

Explain contact forces. Describe muscular force and friction with examples. 5

20. Describe non-contact forces. Explain magnetic, electrostatic and gravitational forces with examples. 5

OR

What is the difference between mass and weight? Calculate the weight of a 5 kg body on Earth and on Jupiter (25.4 N/kg). 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND